Coverage-controlled learning of connection-invariant magnetic surrogates for reconfigurable electric machines

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Abstract

Reconfigurable windings (series–parallel, star–delta, taps, open-end) extend the speed range and part-load efficiency of electric machines, yet every candidate connection is still evaluated with its own dedicated finite-element (FE) model or flux-map table, leaving the combinatorial connection space unexplorable at design time. We treat a winding connection as linear constraints on coil currents, $\boldsymbol{i}_c = \boldsymbol{C}\boldsymbol{i}_p$, and train one connection-invariant surrogate on coil-level FE data. The zero-shot problem is characterised by current-space coverage; its measured rank correlation with flux error is 0.82 over seven hold-outs. A structured co-energy network preserves reciprocity and the torque–flux Maxwell relation with a local residual of 0.67 %, versus 218 % for a black-box MLP. On a seven-connection benchmark, the choice of sampled current patterns dominates sample count: 24 phase-random groups reduce an open-phase hold-out from flux nRMSE 1.25 to 0.088 at one quarter of the cost of 48 magnitude-random groups, while magnitude coverage reduces a half-winding tap from 1.60 to 0.31. Both trends replicate on a 6-slot/4-pole machine. An independent FEMM check gives 8.0 % mean PM fundamental error, 7.5 % saturated-torque error, and end-to-end torque $R^2 = 0.89$–0.92 for three unseen synchronous connections. In a 540-kg single-motor WLTC class-1 case, every fixed connection completes the cycle; DP energy is 0.006 % higher than fixed 4S. A 105-point bus-voltage and gear-ratio sweep finds no case in which reconfiguration supplies feasibility absent from a fixed connection; 24 of 86 feasible cases yield a 1 % or larger energy gain (maximum 2.77 %). Reconfiguration is therefore a modest, regime-dependent

design option rather than a universal efficiency strategy.

Keywords: winding reconfiguration, magnetic co-energy, surrogate modelling, zero-shot generalisation, physics-informed neural networks, drive-cycle optimisation

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1. Introduction

1.1. Motivation

Reconfigurable machine windings—series–parallel changeover, star–delta switching, mid-winding taps, open-end windings and post-fault reconfiguration—trade voltage class against current class at run time. Recent reviews [1, 23] report torque and power-density gains of 40–80 %, machine loss reductions of 30–60 %, and inverter current reductions above 40 % on a production passenger car [1]. The enabling physics has been demonstrated repeatedly: series connection multiplies the back-EMF and suits low-speed high-torque operation, while parallel connection divides it and extends the constant-power speed range [2, 3].

What has *not* kept pace is the modelling machinery. Every published study evaluates two to four hand-picked connections, each with its own FE model or its own dq flux-map table. For a machine with four coil groups per phase the discrete connection space already contains hundreds of meaningful entries (series–parallel groupings, taps, star/delta, open-winding, degraded states), and each FE campaign takes hours to days. The 2025 review is explicit: the field lacks (i) early-stage design methods that compare connections systematically, (ii) drive-cycle-level switching strategies, and (iii) system-level loss accounting that includes the changeover circuit [1]. Concurrently, the transient-control side of the problem has been pushed to near-optimal territory—recent deadbeat predictive control achieves 5 μs seamless changeover with no torque interruption [4, 5, 6]—so the bottleneck has decisively shifted from "how to switch" to "what to switch to, and when".

1.2. Related work and the gap

Analytic connection selection. Normalised characteristic-current criteria map connection families onto a single $\hat{I}_{ch}$–ξ plane and give closed-form decision thresholds [7]. These are valuable screening tools, but they are derived under the assumptions of linearity, negligible cross-saturation and negligible losses; the same paper lists drive-cycle optimisation and switching transients as open problems. Analytic criteria cannot exploit the nonlinear multi-coil physics that FE already provides.

Learned magnetic surrogates. Neural mappings of flux linkage and torque are standard for fixed windings: ANN torque surrogates [8], physics-informed networks [18] including PINN-parameterised PMSM fields [9], POD/SVR fast surrogates [10], Fourier neural operators [19] for FE acceleration [11], and data-efficient *gradient networks* that learn the magnetic co-energy/flux-linkage map with reciprocity and energy conservation enforced by the architecture [12]. All of these operate on the d–q plane of *one* fixed winding; changing the connection changes the input dimension itself, so the model must be retrained. None of them addresses topology generalisation.

Winding-level reconfiguration studies. Series–parallel changeover hardware and control are mature [2, 13, 14]; open-end and multi-mode drives select the best mode pointwise [15]; three-speed tapped IPMSMs [16] and reconfigurable SRM windings [17] extend the principle. Mode selection, however, is either a fixed speed threshold or a per-point greedy choice; no study optimises the *sequence* of modes over a drive cycle with changeover losses and dwell constraints, and losses of the changeover circuit itself are usually ignored [1].

The gap. Three communities hold one piece each: connection physics (electrical machines), structure-enforcing ML (co-energy/gradient networks), and scheduling (energy management). What is missing is the bridge: *a single magnetic model that can be queried across candidate connections, with coverage characterising when transfer is reliable.* If the connection is only a linear constraint

on coil currents, such a model turns connection exploration from "one FE campaign per connection" into "one training run plus a matrix product", and makes drive-cycle co-design of topology and schedule computationally trivial.

1.3. Contributions

1. **Problem formulation and theory (§2).** We cast winding reconfiguration as constrained regression in coil-current space, prove that the co-energy and its derivatives are connection-invariant (Prop. 1), and give a qualitative bound for zero-shot transfer to unseen connections governed by the coverage distance of the training current distribution (Prop. 2). The qualitative implication is consistent with the experiments: connections whose current vectors lie inside the training support transfer at in-distribution accuracy, while support-extrapolating connections fail until the support is extended; the coverage distance correlates with zero-shot error at rank $\rho = 0.82$ over seven hold-outs, with informative misalignments that delimit when the screen applies.
2. **Connection-invariant structured co-energy network (§3).** A scalar co-energy head with a Fourier-parameterised, PSD-seeded inductance matrix, harmonic PM flux and a zero-initialised residual MLP. Reciprocity, energy conservation and torque–flux consistency hold by construction to autograd precision (1.1×10^{-4} reciprocity residual and 0.67 % mixed-partial residual), versus 0.211 and 218 % for a same-size MLP. The structure is most accurate on the hard unbalanced, tap and rotor-only hold-outs, while the MLP retains a small advantage on in-support flux. We also report an instructive negative result: a monotone (PSD-Jacobian, input-convex-style [20]) vector-field parameterisation lags at all difficulties because the joint co-energy is not convex in $(\boldsymbol{i}_c, \theta)$.
3. **A stratified zero-shot benchmark with data-design principles (§4–5).** Seven hold-out connections across three difficulty tiers, 9,276 nonlinear FE samples (23,000+ including the drive-cycle pools), four model families, and a reproducible split protocol. Two findings generalise be-

yond this machine: (i) 48 magnitude-random current patterns reduce the unbalanced hold-out from 0.80 to 0.10 (co-energy net); (ii) phase-random patterns (a quarter of the magnitude family's FE cost) rescue open-phase connections that magnitude patterns cannot (1.25→0.09). Coverage *kind* dominates coverage *amount*; both principles replicate on a 6-slot/4-pole second machine.

4. **Drive-cycle stress test and zero-shot limits (§6).** The surrogate generates mode-conditioned loss maps in milliseconds; a dynamic program over the official WLTC class-1 trace jointly selects connection and operating point under voltage envelopes, current limits, minimum dwell and changeover energy. In the final single-motor calibration, every fixed connection completes the cycle and DP energy is 0.006 % higher. A 105-point voltage/gear sweep finds no feasibility window created by reconfiguration, but 24 feasible designs show 1–2.77 % energy gains. The same surrogate transfers torque to unseen 3S1P, 4P and tap50 synchronous trajectories with end-to-end FEMM $R^2 = 0.89$–0.92 over 36 points each. The drive example therefore exposes a genuine negative result: connection flexibility does not automatically produce an energy or feasibility benefit.

Data, code, trained models and the benchmark protocol are released; all numbers in this paper are reproducible with one command per figure.

2. Problem formulation

2.1. Notation

Consider a machine with N_c independently excitable slot coils (here $N_c = 12$), rotor position θ, coil currents $\boldsymbol{i}_c \in \mathbb{R}^{N_c}$, and phase/port currents $\boldsymbol{i}_p \in \mathbb{R}^{N_p}$. A *winding connection* is a linear map

$$\boldsymbol{i}_c \;=\; \boldsymbol{C}\,\boldsymbol{i}_p, \qquad \boldsymbol{C} \in \mathbb{R}^{N_c \times N_p}, \tag{1}$$

whose entries encode turns, polarity and group membership. Series–parallel groupings, star/delta (with delta line currents absorbed into $\boldsymbol{C}$), taps (zero

columns for idle coils), open-end windings and open-phase degradation are all instances of (1). The seven connections used in this paper, together with the structure of their $\boldsymbol{C}$, are as follows: 4S (series, $\boldsymbol{C}$ one column of ± 1 per phase), 2S2P and 4P (2 and 4 parallel branches, coil currents scaled by 1/2 and 1/4), delta (winding currents from line currents through the standard 1/3-transform), asym (3+1/2+2/1+3 series–parallel splits with inverse-resistance branch sharing), tap50 (half the coils idle: zero columns), and openB (phase-B column removed, isolated-neutral constraint $i_A + i_C = 0$ folded into the remaining columns).

2.2. Connection invariance of the co-energy

Proposition 1 (Connection invariance)**.** *Fix the geometry and materials. Let $W'(\theta, \boldsymbol{i}_c)$ be the magnetic co-energy of the nonlinear magnetostatic problem. Then for every connection $\boldsymbol{C}$ and every admissible port current $\boldsymbol{i}_p$ with $\boldsymbol{i}_c = \boldsymbol{C}\boldsymbol{i}_p$:*

$$\boldsymbol{\psi}_c = \frac{\partial W'}{\partial \boldsymbol{i}_c}, \qquad T = \frac{\partial W'}{\partial \theta}, \qquad \boldsymbol{\psi}_p = \boldsymbol{C}^\top \boldsymbol{\psi}_c, \tag{2}$$

and the triple $(\boldsymbol{\psi}_c, T, W')$ is independent of $\boldsymbol{C}$.

Proof. The magnetostatic problem is posed on coil currents only; $\boldsymbol{C}$ enters exclusely through (1), so W' is evaluated at the same point of the same function regardless of how that point is parametrised by ports. The port flux follows from power invariance: the magnetic power at the ports equals the co-energy rate, $\boldsymbol{\psi}_p^\top \dot{\boldsymbol{i}}_p = \boldsymbol{\psi}_c^\top \dot{\boldsymbol{i}}_c = \boldsymbol{\psi}_c^\top \boldsymbol{C} \dot{\boldsymbol{i}}_p$ for all $\dot{\boldsymbol{i}}_p$, hence $\boldsymbol{\psi}_p = \boldsymbol{C}^\top \boldsymbol{\psi}_c$. Torque is the θ-partial of the same scalar. (Numerical verification on the FE solver: reciprocity of the differential inductance matrix to 3.2×10^{-4} and path independence of $\int \boldsymbol{\psi}_c \cdot d\boldsymbol{i}_c$ to 2.2 %, Table 1.) □

Proposition 1 is elementary, but its computational consequence is the point of this paper: *a surrogate of $(\theta, \boldsymbol{i}_c) \mapsto (\boldsymbol{\psi}_c, T)$ trained once is a surrogate for every connection simultaneously.* Evaluating a new connection costs a matrix product and the solution of a small port-level circuit model, not a new FE campaign.

2.3. What zero-shot transfer costs

Training data live on a union of subspaces $\mathcal{S} = \{\boldsymbol{i}_c = \boldsymbol{C}_k \boldsymbol{i}_p : \boldsymbol{i}_p \in \mathcal{I}_k\}_{k=1}^{K}$ determined by the K training connections and their current ranges $\mathcal{I}_k$. A hold-out connection $\boldsymbol{C}_*$ queries the model along a different subspace $\mathcal{S}_*$. Let $d(\boldsymbol{i}_c, \mathcal{S}) = \min_{\boldsymbol{s}\in\mathcal{S}} \|\boldsymbol{i}_c - \boldsymbol{s}\|$ and let L_ψ be a Lipschitz constant of the true flux map.

Proposition 2 (Coverage bound)**.** *Suppose $\|\hat{\boldsymbol{\psi}}_c(\theta, \boldsymbol{i}_c) - \boldsymbol{\psi}_c(\theta, \boldsymbol{i}_c)\| \le \varepsilon$ for all $(\theta, \boldsymbol{i}_c)$ with $\boldsymbol{i}_c \in \mathcal{S}$ (training-domain accuracy), and both $\hat{\boldsymbol{\psi}}_c$ and $\boldsymbol{\psi}_c$ are L-Lipschitz in $\boldsymbol{i}_c$ on a convex hull containing $\mathcal{S} \cup \mathcal{S}_*$. Then for every $\boldsymbol{i}_{c,*} \in \mathcal{S}_*$,*

$$\|\hat{\boldsymbol{\psi}}_c(\theta, \boldsymbol{i}_{c,*}) - \boldsymbol{\psi}_c(\theta, \boldsymbol{i}_{c,*})\| \;\le\; \varepsilon + (L + L_{\hat{\psi}})\, d(\boldsymbol{i}_{c,*}, \mathcal{S}).$$

The same statement holds port-wise with an extra factor $\|\boldsymbol{C}_*\|_2$ for $\boldsymbol{\psi}_p = \boldsymbol{C}_*^\top \boldsymbol{\psi}_c$.

Proof. Pick $\boldsymbol{s} \in \mathcal{S}$ attaining the distance. Two triangle inequalities give $\|\hat{\boldsymbol{\psi}}_c(\boldsymbol{i}_{c,*}) - \boldsymbol{\psi}_c(\boldsymbol{i}_{c,*})\| \le \|\hat{\boldsymbol{\psi}}_c(\boldsymbol{i}_{c,*}) - \hat{\boldsymbol{\psi}}_c(\boldsymbol{s})\| + \|\hat{\boldsymbol{\psi}}_c(\boldsymbol{s}) - \boldsymbol{\psi}_c(\boldsymbol{s})\| + \|\boldsymbol{\psi}_c(\boldsymbol{s}) - \boldsymbol{\psi}_c(\boldsymbol{i}_{c,*})\| \le L_{\hat{\psi}} d + \varepsilon + L d.$ □

The bound is qualitative: neither L nor $L_{\hat{\psi}}$ is estimated, so it predicts ordering tendencies, not numeric error. Its value is the screening logic—distance to the support is the only channel through which a connection can hurt—which §5 evaluates empirically with the measured $\bar{d}$.

Remark 1. *The bound contains no term for the* label *of the connection—only the geometric distance of its current vectors to the training support. It is consistent with the experimental stratification of §5: scaled copies of a training subspace (4P of 4S) transfer perfectly; phase-shifted combinations (delta) transfer partially; unbalanced current ratios (asym) transfer only after magnitude-random patterns shrink d; and open-phase patterns, which contain current vectors with sign-flipped phase pairs that magnitude perturbations never produce, require* phase-*random patterns. Section 5 computes $\bar{d}$ per hold-out and reports the rank correlation. The bound also explains why more of the wrong data does not help: d is a property of the support geometry, not of the sample density.*

2.4. Learning problem

Given samples $\{(\theta_j, \boldsymbol{i}_c^{(j)}, \boldsymbol{\psi}_c^{(j)}, T^{(j)})\}_{j=1}^N$ on $\mathcal{S}$, learn $\hat{\boldsymbol{\psi}}_c, \hat{T}$ (or a potential $\hat{W}'$) minimising held-out error on $\mathcal{S}$, and evaluate *zero-shot* on subspaces $\mathcal{S}_* \not\subseteq \mathcal{S}$. Model selection uses only in-support validation data; hold-out connections are touched once, at reporting time. This protocol is stricter than the common practice of early-stopping on the test distribution and is part of the released benchmark.

3. Connection-invariant surrogates

All models share the input $x = [\boldsymbol{i}_c/60,\ \phi(\theta)] \in \mathbb{R}^{19}$ with harmonic features $\phi = [1, \sin 2\theta, \cos 2\theta, \sin 4\theta, \cos 4\theta, \sin 6\theta, \cos 6\theta]$ (electrical harmonics of the 4-pole machine), and predict the 13 physical outputs $(\boldsymbol{\psi}_c, T)$.

3.1. Structured co-energy network (ours)

For potential-based models the network represents $\hat{W}'(\boldsymbol{i}_c, \phi) = I_0 \hat{w}(\boldsymbol{\xi}, \phi)$, $\boldsymbol{\xi} = \boldsymbol{i}_c/I_0$, with $I_0 = 60\,\mathrm{A}$, so current normalisation does not change the physical torque or the Maxwell relation. The scalar head is

$$\hat{w}(\boldsymbol{\xi}, \phi) = \tfrac{1}{2}\, \boldsymbol{\xi}^\top \mathbf{L}(\phi)\, \boldsymbol{\xi} + \boldsymbol{p}(\phi)^\top \boldsymbol{\xi} + R_\theta(\boldsymbol{\xi}, \phi), \tag{3}$$

with the three terms parameterised as follows.

Inductance matrix. $\mathbf{L}(\phi) = \mathbf{A}\mathbf{A}^\top + \epsilon\mathbf{I} + \sum_{h\in\{4,8\}} \left[\mathbf{C}_h \cos h\theta + \mathbf{S}_h \sin h\theta\right]$, each $\mathbf{C}_h, \mathbf{S}_h$ symmetrised. The $\mathbf{A}\mathbf{A}^\top + \epsilon\mathbf{I}$ seed makes the current-independent part positive definite, so at initialisation the model reduces to a stable linear-inductance model; the angle harmonics (orders 4 and 8, i.e. the slotting harmonics of the 12-slot/4-pole geometry) modulate it. Cos/sin of 8θ are composed exactly from the ϕ columns via double-angle identities, keeping a single autograd path.

PM flux. $\boldsymbol{p}(\phi) = \mathbf{P}\phi$, a full harmonic PM flux linkage per coil (orders 2, 4, 6 plus offset).

Saturation residual. R_θ is an MLP on $(\boldsymbol{i}_c, \phi)$ whose final layer is *zero-initialised*. Training therefore starts from the closed-form linear+PM model and

grows a correction only where the data demand it—the same spirit as physics-model-plus-residual learning, but with the residual inside a scalar potential.

Flux and torque are obtained by automatic differentiation,

$$\hat{\boldsymbol{\psi}}_c = \nabla_{\boldsymbol{i}_c} \hat{W}', \qquad \hat{T} = I_0 \, \nabla_\phi \hat{w} \cdot \frac{d\phi}{d\theta}, \tag{4}$$

so both outputs are exact gradients of one scalar. The training loss normalises each output by its training standard deviation only inside the loss; potential-based predictions remain in physical units. It up-weights the single torque channel ($w_T = 3$) to compensate for its 1/13 gradient share.

Proposition 3 (Structural consistency)**.** *For* (3)–(4)*, (i) the Jacobian* $\partial \hat{\boldsymbol{\psi}}_c / \partial \boldsymbol{i}_c = \mathbf{L}(\phi) + \nabla^2_{\boldsymbol{i}_c} R_\theta$ *is symmetric (reciprocity) exactly; (ii) the mixed identity* $\partial \hat{\psi}_k / \partial \theta = \partial \hat{T} / \partial i_k$ *holds exactly (torque–flux consistency); (iii)* $\hat{W}'$ *is a well-defined scalar potential, so no operating point can exhibit non-conservative flux–torque behaviour in downstream optimisation.*

Proof. (i) is the symmetry of the Hessian of a twice-differentiable scalar; (ii) is the commutation of mixed partials; (iii) is immediate. □

3.2. Baselines

Direct MLP. A 3-layer width-128 GELU network regressing the 13 outputs: the accuracy ceiling without structure. **Per-connection MLP.** The industrial status quo: a direct MLP trained only on the nominal connection's data (4S), reported on hold-out connections. **Monotone vector field (negative result).** The architecture of [12] generalised to 12 coils: $g(x) = \mathbf{A}_0 x + \boldsymbol{b}_0 + \mathbf{A}_1^\top \sigma(\mathbf{A}_1 x + \boldsymbol{c})$ with σ = softplus and $\mathbf{A}_0 = \mathrm{blkdiag}(\mathbf{B}^\top \mathbf{B} + \epsilon \mathbf{I}, 0)$, which makes the field g the exact gradient of a convex scalar and its Jacobian symmetric positive semidefinite everywhere. Despite perfect reciprocity (1.6×10^{-5}, Table 6), it cannot fit this machine at any difficulty tier (§5.4). **Reciprocity-structured ridge.** Linear-in-current model $\hat{\psi}_i = \sum_h H_h(\theta) \big[a_{ih} + \sum_{j \le k} S_{ijk} \, m_{jk}(\boldsymbol{i}_c) \big]$ with symmetric S—the "linear physics" reference.

3.3. Training protocol

Full-batch AdamW, 10^4–1.2×10^4 epochs, gradient clipping, and early stopping on a *stratified validation slice of the training connections only* (12 %, never on hold-out rows). Three seeds were used for the learning-curve experiment; the fixed-seed benchmark uses one run per model and split. A single training run takes tens to several hundred seconds on a laptop GPU; batched inference costs tens of microseconds per point on GPU, roughly five orders of magnitude faster than the 1.2 s FE reference solve; the total compute behind every number in this paper is under six GPU-hours plus 80 CPU-core-hours of FE solving.

4. FE data and benchmark protocol

4.1. Machine and solver

The case-study machine is a 12-slot/4-pole surface-mounted PM machine, solved with an in-house 2-D nonlinear magnetostatic solver: Fröhlich-type iron $\mu_r(B)$, radial PMs, one independent current per slot, coil flux linkage from the vector potential, and torque from the derivative of the discrete magnetic co-energy. The permanent-magnet annulus has rotating 80 % pole arcs; the inter-pole gaps are non-magnetic. The training mesh is 40×96 (≈ 1.2 s per nonlinear solve, 8-way parallel); an 80×192 re-solve of 64 probe points quantifies discretisation uncertainty (Table 1). Machine parameters: 2 pole pairs of surface PM (radial, B_r-equivalent magnetisation 1.2 T), stack length 100 mm, 40 turns per coil, slot-pitch occupancy 0.52 and homogenised slot current; the drive resistance uses a 40 % copper fill. The iron model is Fröhlich ($\mu_{r0} = 4000$, $B_{sat} = 1.6$ T); per-coil resistance 2.4 mΩ from geometry. We deliberately use a research-grade solver: the claims of this paper concern relative model behaviour against a fixed reference, not absolute machine performance.

To make the torque convention explicit, let

$$W'(\boldsymbol{i}_c, \theta) = L_{\text{stack}} \int_\Omega A_z J_z \, \mathrm{d}S - L_{\text{stack}} \int_\Omega \int_0^B \nu(b) b \, \mathrm{d}b \, \mathrm{d}S. \tag{5}$$

Torque is evaluated by a $\pm 7.5^\circ$ central difference of W' at fixed currents, and the same treatment is used for the zero-current PM baseline. The stack length

therefore enters both flux linkage and torque. The verification target is the discrete Maxwell relation $\partial T/\partial i_k = \partial \psi_k/\partial \theta$.

4.2. Datasets

All datasets share the rotor grid $\theta \in \{0, \ldots, 165°\}$, step 15°. The **zero-shot benchmark pool** (9,276 samples) consists of the first, second, third and last bullets below; the phase-random family enters the coverage ablations of §5.2, and the last two bullets are generated exclusively for the drive-cycle study:

- **Standard connections** (1,344): 4S, 2S2P, delta, asym at 7×4 current-amplitude/angle combinations.
- **Magnitude-random patterns** (4,608): base 4S excitation, each coil scaled by $U(0.2, 1)$, RMS-renormalised; 48 groups.
- **Zero-support patterns** (2,304): scale $U(0, 1)$, for the tap/fault ablation.
- **Phase-random patterns** (1,152): per-phase amplitude $U(0.2, 1)$, per-phase angle $U(0, 2\pi)$, plus zero-sequence $U(-0.3, 0.3)$—current vectors with arbitrary phase relations.
- **Dense current-angle sweep** (2,304): 4S/2S2P/delta/4P over 12 current angles per 120° sector.
- **Synchronous-path set** (10,368): the four switchable modes along their actual synchronous trajectories $i_p(\theta) = I[\cos(2\theta + \gamma), \ldots]$ with γ over the full 360° electrical domain (36 values) and $I \in \{10, \ldots, 60\}$ A—the distribution the drive-cycle study queries. (The torque–current-angle characteristic of this machine has no sub-360° period: we verified with dedicated FE solves that, once saturation is active, neither the three-phase permutation symmetry nor the half-period sign symmetry holds exactly, so the full domain must be sampled.)
- **Zero-current PM set** (12) and **hold-out connections** (1,008): 4P (quarter-scaled copy of 4S), tap50 (half-winding series tap), openB (phase-B open, isolated neutral, $i_A + i_C = 0$).

Table 1: FE solver consistency checks and discretisation uncertainty.

Check	Result	Interpretation
Linear mutual-inductance reciprocity	3.0×10^{-6}	solver-level
Nonlinear differential reciprocity (20 A bias)	3.2×10^{-4}	solver-level
Co-energy path independence (2 paths)	2.2 %	solver-level
Maxwell check, 12-slot ($\partial T/\partial i$ vs. $\partial \psi/\partial \theta$)	$r = 0.99998$, 2.4 %	solver-level
Maxwell check, 6-slot	$r = 0.9998$, 2.3 %	solver-level
Flux, 80×192 vs 40×96	$r = 0.997$, ratio 1.06	discretisation
Torque, 80×192 vs 40×96	$r = 0.751$, $R^2 = 0.377$	discretisation

4.3. Protocol

Two evaluation modes. **LOCO** (leave-one-connection-out): train on the other three standard connections plus patterns, test on the hold-out one. **EXT**: train on all four standard connections + patterns, test on 4P/tap50/openB which are never seen in any form. Three difficulty tiers follow from Prop. 2: *tier 1* in-support scaling (2S2P, 4P), *tier 2* phase-recombination (delta), *tier 3* support-extension (asym: unbalanced ratios; tap50/openB: zero/excluded coils and sign-flipped phase pairs). Metrics: flux nRMSE (per-coil RMSE normalised by per-coil test standard deviation, averaged over 12 coils) and torque R^2.

Table 2 reports an independent FEMM 4.2 cross-check of the same geometry and material curve, with 40 turns per coil, a 100-mm stack and the same BH points. The linear inductance matrix has correlation 0.999996 with a 120×288 in-house reference; after one global factor of 1.13, its shape residual is 2.1 %. The remaining global factor is not calibrated away: it indicates that the two solvers do not yet share an identical absolute normalisation. After correcting the rotating 80 % PM arc, the PM-only waveform has correlation 0.9993 and a fundamental-amplitude error of 7.9 % on average (8.4 % maximum), while a saturated 4S case reaches flux correlation 0.9992 with a mean pointwise flux difference of 20.2 % and a 7.5 % torque difference. The absolute flux scale still precludes claims of pointwise accuracy. The cross-check also tests zero-shot screening. We retrained with the zero-support "withz" pool and evaluated 36

Table 2: Independent FEMM 4.2 cross-check. The inductance discrepancy is computed after one global scale factor; nonlinear flux is pointwise and torque uses the weighted-stress integral.

Comparison	Correlation	Discrepancy
Linear inductance, fine reference	0.999996	2.1 % after 1.13×
PM-only flux waveform	0.9993	7.9 % fundamental (max 8.4 %)
Nonlinear 4S flux (60 A)	0.9992	20.2 % mean pointwise
Nonlinear 4S torque (60 A)	—	7.5 %

Table 3: Three-way zero-shot validation at 36 synchronous points. 4P/tap50 use the zero-support benchmark MLP; 3S1P uses the drive-cycle MLP trained on synchronous paths. Entries are correlation / torque R^2 (MAE in N m). "Proxy–reference" isolates generalisation under the training solver; "reference–FEMM" isolates the solver difference; "proxy–FEMM" is the end-to-end error.

Hold-out	Flux corr. (proxy–ref.)	Torque proxy–ref.	Torque ref.–FEMM	Torque proxy–FEMM
4P	0.771	0.983 / 0.959 (0.357)	0.988 / 0.951 (0.498)	0.972 / 0.918 (0.697)
tap50	0.580	0.957 / 0.835 (1.428)	0.994 / 0.975 (0.721)	0.958 / 0.887 (1.389)
3S1P	0.940	0.952 / 0.857 (1.265)	0.995 / 0.967 (0.762)	0.952 / 0.894 (1.274)

independent synchronous points. The zero-support MLP is used for 4P and tap50; the drive-cycle MLP, which has not seen 3S1P, is used for that connection. Proxy-versus-reference and solver-versus-solver errors are reported separately in Table 3. The never-trained 3S1P connection reaches end-to-end torque $R^2 = 0.895$ against FEMM; the 4P and tap50 connections reach 0.918 and 0.887. Their torque correlations are all 0.95–0.97. The flux channel is less transferable for the direct MLP (proxy–reference correlations 0.58–0.77), whereas the co-energy model raises 4P/tap50 to 0.954/0.903. Zero-shot torque screening is therefore useful for prioritizing FE verification, but an absolute flux or loss decision still requires an independent solve.

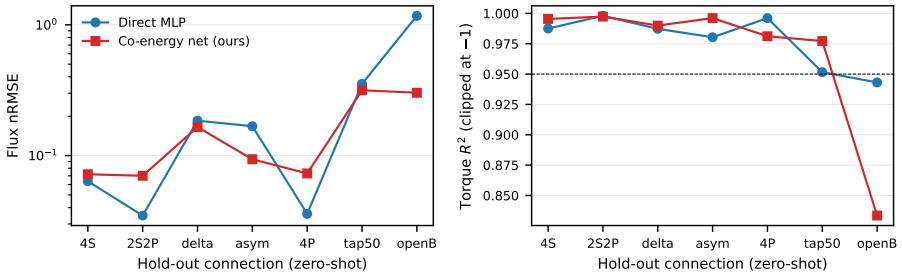


Figure 1: Zero-shot benchmark across the seven hold-out connections (base variant). Left: flux nRMSE (log scale); right: torque R^2. Tier-1 hold-outs transfer at in-distribution accuracy; tier-3 hold-outs require pattern coverage; the monotone vector field lags on every hold-out.

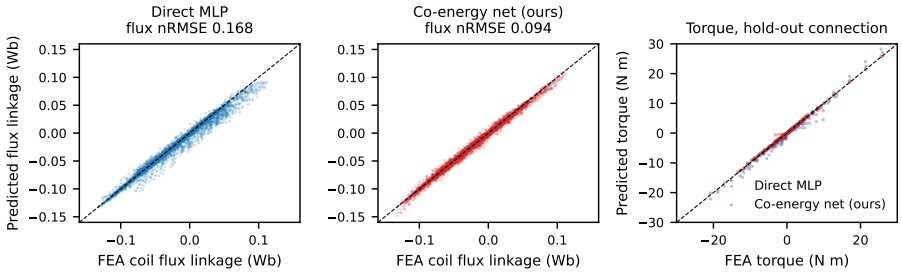


Figure 2: Hold-out connection *asym*: predicted vs FE coil flux linkages (left: direct MLP; centre: co-energy net) and torque (right).

5. Zero-shot results

5.1. Main benchmark

Table 4 and Fig. 1 report all hold-outs; Fig. 2 shows parity plots for the hardest LOCO hold-out. As a quantitative anchor for Prop. 2, we compute for each hold-out the mean min-distance $\bar{d}$ of its test current vectors (normalised by the 60 A scale) to the sampled training support: 4P 0.036, 2S2P 0.055, asym 0.206, delta 0.240, 4S 0.263, openB 0.288, tap50 0.536. Across the seven hold-outs the rank correlation with the zero-shot flux error is $\rho = 0.82$ ($p = 0.02$). The correlation is meaningful but not perfect, and the misalignment is informative: openB sits at a modest $\bar{d}$ yet the base MLP remains at 1.17, because its sign-flipped phase pairs are *close* in Euclidean distance to the training support while

lying in a direction the support barely samples—an unweighted distance metric cannot see support density or direction. The 4S hold-out supplies the opposite counterexample: its distance is high (0.263) while its flux error is low (0.064). We therefore use $\bar{d}$ only as a coarse risk indicator; neither a large nor a small value justifies accepting or rejecting a connection. Four observations.

(1) In-support transfer is essentially free. For tier-1 hold-outs (2S2P, 4P) the shared model reaches flux nRMSE 0.035–0.073 and torque $R^2 \geq 0.981$, i.e. in-distribution accuracy: their current subspaces lie inside the training support ($\bar{d} < 0.06$, Prop. 2), so the connection adds no error of its own. Notably, 4P was *never seen*—it is a rescaled 4S, and the model treats it as such.

(2) Without pattern coverage, tier-3 collapses. When trained on the standard connections only ("no-random"), the MLP and co-energy network predict asym flux at nRMSE 0.866 and 0.803, with torque $R^2 \approx 0.50$; both channels lose the cross-current structure. The monotone vector field is worse still. This is the single most important negative control in the benchmark: connection labels are worthless; current-space coverage is everything.

(3) Magnitude patterns rescue unbalance. Adding 48 magnitude-random groups brings asym to 0.168 (MLP) and 0.094 (co-energy net); torque R^2 reaches 0.980 and 0.996, respectively.

(4) Phase patterns rescue open-phase. Magnitude patterns leave openB at 1.17; 24 phase-random groups bring it to 0.068 (§5.2)—the second design principle.

5.2. What kind of coverage matters

Fig. 3 shows the coverage-scaling curves; Table 5 isolates the effect: the two pattern families rescue *disjoint* connection classes. Magnitude patterns bring tap50 from 1.60 to 0.35 (they sample non-uniform per-coil amplitudes, including small ones) but leave openB at 1.37 (open-phase operation drives sign-flipped phase pairs that multiplicative perturbations of a balanced base never generate). Phase-random patterns do the opposite: openB drops to 0.068 ($R^2 = 0.99$) and delta from 0.364 to 0.062 (delta is itself a phase-recombination), while

Table 4: Zero-shot benchmark, flux nRMSE / torque R^2. Training variants: "base" = three standard connections (all four for EXT rows) + 48 magnitude-random groups + PM data; "withz" = base + the 24 zero-support pattern groups; "no random" = standard connections + PM data only. LOCO rows hold out one standard connection; EXT rows (4P, tap50, openB) hold out connections never seen at all. Best per row in bold.

Tier	Hold-out	Direct MLP	Co-energy (ours)	Monotone VF
1	4S (LOCO)	**0.064**/0.988	0.072/**0.995**	0.334/0.893
1	2S2P (LOCO)	**0.035**/**0.998**	0.070/0.997	0.270/0.776
1	4P (EXT)	**0.036**/**0.996**	0.073/0.981	0.288/0.495
2	delta (LOCO)	0.185/0.987	**0.165**/**0.990**	0.309/0.748
3	asym (LOCO)	0.168/0.980	**0.094**/**0.996**	0.483/0.485
3	tap50 (EXT)	0.354/0.952	**0.316**/**0.977**	0.386/0.815
3	openB (EXT)	1.172/**0.943**	**0.302**/0.834	0.334/0.599
asym, no random patterns		0.866/0.493	0.803/0.500	7.89/−2.39
asym, per-connection MLP (4S only)		0.866/0.493	—	—
asym, linear ridge (reciprocity-structured)			0.545	

[†]with zero-support patterns included ("withz"): tap50 flux nRMSE/R^2 becomes 0.284/0.977; openB 0.389/0.866.

tap50 degrades to 1.48 in that pool alone (phase-random patterns keep all coils energised). With both families in the pool (a quarter of the magnitude family's FE cost added on top), each hold-out is served by the family it needs: openB 0.09, delta 0.077, tap50 0.384 (via either family's best). On delta, scaling the magnitude family from 0 to 48 groups only reaches 0.186 (0.364 at zero groups and 0.207 at 24), while a quarter-sized phase family reaches 0.062: coverage *kind* dominates coverage *amount*. This is consistent with the qualitative implication of Prop. 2, although the bound itself is not tight in Euclidean distance.

5.3. Physical consistency

Table 6 and Fig. 4 evaluate the flux Jacobian $\partial\hat{\boldsymbol{\psi}}_c/\partial\boldsymbol{i}_c$ by central differences on 24 held-out operating points. The co-energy network violates reciprocity by 1.1×10^{-4}—over three orders of magnitude below the direct MLP's 0.211—and keeps a positive-definite symmetric part (min eigenvalue $0.028 > 0$). A local mixed-partial probe gives a torque-flux residual of 0.67 % for the co-energy network versus 218 % for the MLP. The monotone vector field is symmetric by

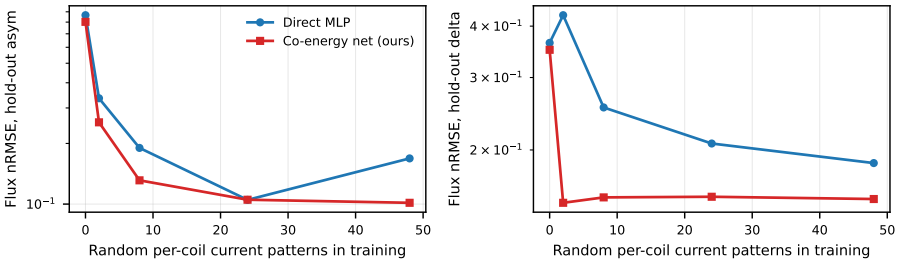


Figure 3: Coverage scaling of magnitude-random patterns on the two hardest standard hold-outs. The co-energy model improves monotonically on asym; delta is insensitive to magnitude coverage and requires phase coverage (Table 5).

Table 5: Coverage-kind ablation (flux nRMSE / torque R^2). Hold-outs: openB = phase-B open circuit, tap50 = half-winding series tap, delta = star–delta recombination. Magnitude patterns (48 groups, 4,608 FE samples) and phase patterns (24 groups, 1,152 samples) rescue *different* hold-out connections; the union covers all.

Training pool	openB	tap50	delta
standard only	1.37/0.37	1.60/0.12	0.364/0.88
+ magnitude (48 groups)	1.17/0.94	**0.35**/0.95	0.186/0.99
+ phase (24 groups)	**0.068**/**0.99**	1.48/0.03	**0.062**/**1.00**
+ both	0.092/0.99	0.384/0.85	0.077/**1.00**

construction (5.2×10^{-5}) but has substantially worse accuracy.

5.4. Negative result: monotonicity is the wrong structure

The monotone vector field enforces joint convexity of the underlying scalar in $(\boldsymbol{i}_c, \phi)$. The true co-energy is approximately convex in $\boldsymbol{i}_c$ alone at fixed θ, but its θ-dependence through the PM field and slotting makes the joint Hessian indefinite; the architecture can therefore only fit a compromise, and lands at flux nRMSE 0.27–0.48 and torque R^2 no higher than 0.89, including in-support (4S LOCO 0.33, single-connection-trained variant 0.42), so the shortfall is not an artefact of cross-connection sharing. Stronger training (2× epochs, 3× learning rate) does not close the gap either: the failure is structural rather than one of optimisation. The lesson for structural design in magnetic ML: *enforce the*

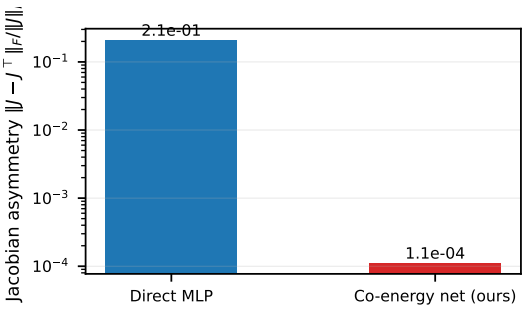


Figure 4: Flux-Jacobian reciprocity residual on held-out operating points (log scale): the co-energy network violates reciprocity nearly three orders of magnitude less than the direct MLP.

Table 6: Jacobian reciprocity, positive-definiteness and local torque–flux residual on held-out asym points.

Model	$\|J - J^\top\|_F / \|J\|_F$	min eig$(\frac{J+J^\top}{2})$	Torque–flux residual
Direct MLP	0.211 (max 0.310)	0.368	218 %
Co-energy (ours)	1.1×10^{-4} (max 3.5×10^{-4})	0.028	0.67 %
Monotone VF	5.2×10^{-5}	0.015	1.4 %

physical symmetries the PDE actually has (reciprocity, potential structure), not convenient convexity properties it does not have. Scalar co-energy parameterisations get the former for free.

5.5. Data efficiency and coverage scaling

Scaling the magnitude-pattern family on the unbalanced hold-out (asym, co-energy net): 0 groups 0.80, 2 groups 0.25, 8 groups 0.13, 24 groups 0.105, 48 groups 0.102—a clean curve that saturates near 24 groups, i.e. a few thousand FE samples. The direct MLP reaches the same range at 24 groups but is non-monotone at 48 (0.17), so more samples of the same family are not uniformly better. On delta the magnitude family saturates near 0.15 while the quarter-sized phase family reaches 0.06 (§5.2), which is the coverage-kind argument in its purest form.

Fig. 5 shows subsampling of the full training pool (three seeds, hold-out

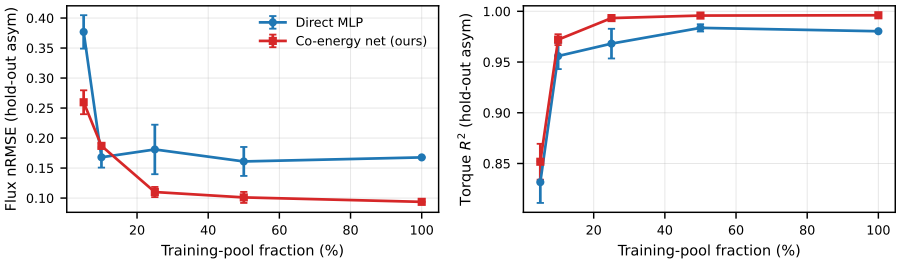


Figure 5: Data-efficiency learning curves on the hold-out connection (three seeds): flux nRMSE (left) and torque R^2 (right) vs fraction of the training pool.

asym): the MLP reaches nRMSE 0.17 with 10 % of the data, while the co-energy net reaches 0.26 at 5 %, 0.19 at 10 %, 0.11 at 25 %, and 0.094 on the full pool. Its torque R^2 is 0.85 even at 5 % and rises to 0.996 on the full pool. The practical reading is therefore conditional: the structured head is competitive at very low data, becomes clearly better after a quarter of the pool, and does so while retaining physical consistency.

5.6. Cross-machine replication

To test whether the two coverage principles are artefacts of one machine, we repeated the core ablations on a second machine with the same solver family: a 6-slot/4-pole SPM (different slot count, winding factor, inductance matrix and harmonic content; 3,084 FE samples in the same dataset families). With a direct MLP and the identical protocol, the pattern repeats: the half-winding tap hold-out improves with *magnitude*-random patterns (flux nRMSE $0.45 \rightarrow 0.21$; phase patterns alone leave it at 0.47), while the open-phase hold-out is rescued by *phase*-random patterns ($0.59 \rightarrow 0.050$; magnitude patterns alone leave it at 0.77). The union pool again serves both (0.21 and 0.045). Which family a connection needs tracks the geometry of its current vectors, not the machine—as the coverage argument requires.

Table 7: Cross-machine replication on the 6-slot/4-pole SPM. Entries are flux nRMSE / torque R^2 for the direct MLP. The co-energy network was not re-trained for this replication, so the table tests the two sampling principles rather than architecture ranking.

Hold-out	Standard	+ magnitude	+ phase	+ both
tap50	0.454/0.705	**0.209**/0.940	0.466/0.548	0.211/**0.973**
openB	0.588/0.862	0.773/0.607	0.050/**0.996**	**0.045**/0.994

6. Drive-cycle co-design case study

6.1. From surrogate to system losses

Unless stated otherwise, the drive-cycle study uses the direct-MLP surrogate as the reference model: it is the most accurate of §5 on the connections involved, and the co-energy surrogate is used as an independent cross-check. The surrogate is queried in a synchronous-rotating frame. For a mode m with connection matrix $\boldsymbol{C}_m$, line-current amplitude I and current angle γ, the phase currents rotate as $i_p(t) = I[\cos(2\theta+\gamma), \cos(2\theta+\gamma-\frac{2\pi}{3}), \cos(2\theta+\gamma+\frac{2\pi}{3})]$. Average torque, copper loss, the iron-loss proxy and the port voltage then follow from one electrical period of surrogate evaluations (12× torque grid, 72× voltage grid), giving loss maps $P^m_{\text{loss}}(T,\omega)$ and voltage envelopes $V^m(I,\omega) \le V_{dc}/\sqrt{3}$ per mode in under a second on GPU. The loss model comprises exact copper loss $\sum_c i_c^2 R_c$ from the coil currents; inverter conduction and switching from declared device parameters ($V_{CE0} = 0.7\,\text{V}$, $r_{CE} = 2\,\text{m}\Omega$, $E_{sw} = 1\,\text{mJ}$ per period at 10 kHz, a declared linear SiC stage); a Bertotti-type [21] iron-loss proxy on the AC coil-flux amplitude (coefficients stated in the released configuration, ±2× swept); and a changeover energy E^c_{sw} per reconfiguration event (5 mJ–100 J swept). The vehicle is a 540 kg single-motor EV driving through gear 13 (wheel 0.25 m, $C_dA = 0.50\,\text{m}^2$, 320 V bus, 40 % slot fill, $R_c = 3.02\,\text{m}\Omega$). This gives fixed 4S a motor-side efficiency of 97.0 % (93.1 % after the 96 % gear efficiency), and the cycle is the official WLTC class-1 trace [22] (JRC data, checksum-verified). The peak machine demand is 9.14 N m at 8,883 rpm; all four fixed connections complete the cycle.

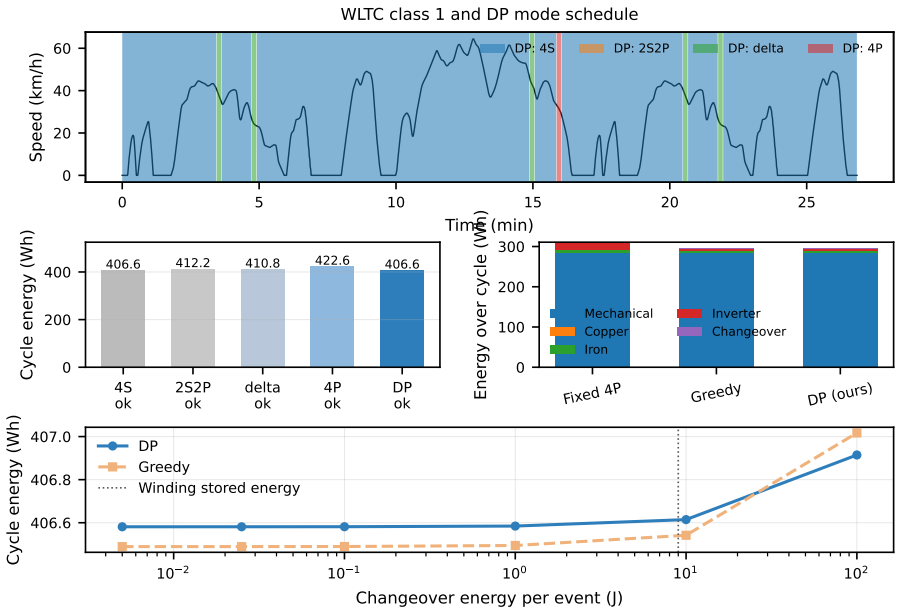


Figure 6: Drive-cycle case study. Top: WLTC class-1 speed trace and the DP mode schedule (colour band). Middle-left: fixed-connection and DP cycle energy; all listed strategies complete the cycle. Middle-right: loss decomposition for the completed trajectories. Bottom: sensitivity of cycle energy to the changeover energy, swept from 5 mJ to 100 J.

6.2. Scheduling formulation

With per-step mode-conditioned optimal operating points, mode selection is a shortest-path problem over a mode graph with dwell constraints. We solve it by dynamic programming on the state (mode, steps-since-last-switch) with minimum dwell $D = 10$ s, comparing: **fixed-4S** (the best fixed connection here), **fixed-4P** (the conservative single-inverter baseline), **2-mode rule** (4S/4P at a swept speed threshold, the practice of [2]), **pointwise greedy**, and **DP**.

6.3. Results

Table 9 and Fig. 6 summarise the case study.

Reconfiguration has negligible energy value in the nominal case. Fixed 4S is the best fixed connection at 406.56 Wh (35.58 Wh/km); 2S2P, delta and 4P complete the same cycle at 412.25, 410.77 and 422.63 Wh. DP uses

406.58 Wh, which is 0.006 % *higher*. Its loss decomposition closes exactly: copper increases by 0.32 Wh, iron falls by 3.70 Wh, inverter loss falls by 12.66 Wh, and mechanical work is unchanged. The changes nearly cancel; there is no topology-driven efficiency improvement.

Scheduling buys switching-burden reduction, not energy. The pointwise greedy sequence uses 406.489 Wh with 19 switches; DP uses 406.581 Wh with 12 switches. DP sacrifices 0.092 Wh for a modest reduction in switching. A swept two-mode rule uses 407.592 Wh with eight switches. The modes used by DP are 4S (1,545 s), delta (55 s), and 4P (11 s); 2S2P is not selected.

The saving is robust across physically meaningful changeover energies. The default per-event cost is 25 mJ, while the winding stores about 9 J at typical commutation currents. Sweeping over 5 mJ–100 J keeps the DP schedule at 12 switches; DP energy changes from 406.581 to 406.9 Wh. The conclusion is insensitive to a physical changeover-energy range because there is no energy benefit to preserve.

Surrogate and device sensitivity. Re-evaluating the same schedule with the co-energy surrogate gives a -0.022 % saving relative to its own best fixed connection. Halving r_{CE} with V_{CE0} increased by 10 % gives -0.006 %. Both checks agree with the direct MLP: the result is economically null.

6.4. Zero-shot modes and connection screening

The drive study above schedules four connections that all appear in the training pools. The point of the connection-invariant formulation, however, is that a candidate $\boldsymbol{C}$ can be queried without new FE data or retraining, subject to coverage. Two experiments exercise this.

A zero-shot candidate is screened and independently checked. We add tap50—never present in the drive surrogate's training pool in any form—as a fifth switchable mode. The five-mode DP assigns it 11 s and reduces energy from 406.581 to 406.566 Wh, a negligible 0.004 % gain. The independent FEMM trajectory gives torque $R^2 = 0.887$ for the direct MLP and 0.908 for the co-energy model, supporting the predicted marginal role. The never-trained 3S1P

Table 8: Exploratory zero-shot screening: fixed-connection cycle energy and unmet torque integral (WLTC class 1). Entries marked * are absent from the drive training pool. These rankings are hypotheses; missed operating points still require targeted FE or measurement validation.

Connection	Energy (Wh)	Unmet torque (N m s)
4S	406.6	0.0
2S2P	412.2	0.0
delta	410.8	0.0
4P	422.6	0.0
3S1P*	423.6	0.0
tap75*	(342.1)	135.2
tap50*	416.4	0.0
tap25*	(455.3)	215.0

is also credible ($R^2 = 0.895$) but consumes 412.1 Wh as a fixed connection, so it is not selected. Zero-shot torque screening is quantitatively useful, while flux and loss remain subject to the absolute-scale limitation.

Connection screening at design time. Table 8 evaluates eight candidate connections (including three tap ratios and an unbalanced 3S1P split, all zero-shot for the tap/3S1P entries) as fixed connections over the same cycle. All eight entries except tap75 and tap25 complete the cycle; their energy ranking still requires targeted FE checks before a design decision, but the independent tap50 check supports the screening direction.

No reconfiguration-required window in the tested design grid. Fig. 7 evaluates 105 combinations of three cycles, seven bus voltages (180–700 V) and five gear ratios (8–20). Sixty-two have at least one fully feasible fixed connection without a 1 % gain, 24 gain at least 1 % from DP, and 19 are infeasible for both fixed and reconfigurable operation. No point is completed by DP while every fixed connection fails. Among the 86 feasible cases, the median DP gain is 0.479 % and the maximum is 2.77 %. The gains concentrate in low-voltage or high-speed class-2/3 operating regions; the nominal class-1

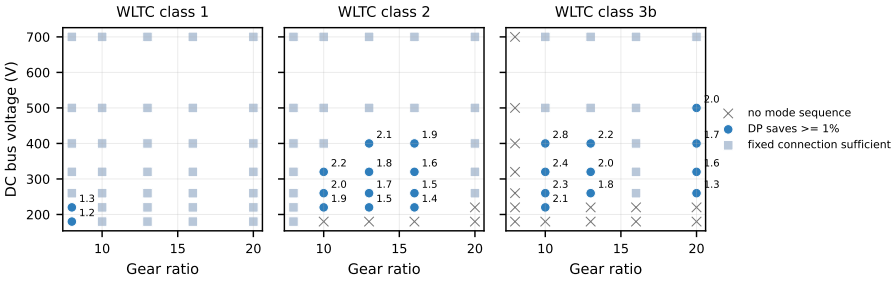


Figure 7: Bus-voltage/gear sweep over WLTC classes 1, 2 and 3b. Circles are cases in which DP saves at least 1 %; squares are feasible fixed connections with negligible gain; crosses are infeasible even with all four modes. No feasibility-only case occurs; the circles mark the modest energy-benefit regime.

Table 9: Drive-cycle case study: WLTC class-1, 540-kg single-motor EV, 320-V SiC drive, gear 13, 11.43 km, all modelled losses counted, minimum dwell 10 s. All strategies complete the cycle.

Strategy	Complete	Energy (Wh)	Wh/km	vs DP	Switches	Unmet torque (N m s)
Fixed 4S	yes	406.56	35.58	−0.006 %	0	0
Fixed 2S2P	yes	412.25	36.08	+1.375 %	0	0
Fixed delta	yes	410.77	35.95	+1.038 %	0	0
Fixed 4P	yes	422.63	36.98	+3.794 %	0	0
2-mode rule	yes	407.59	35.67	+0.248 %	8	0
Pointwise greedy	yes	406.49	35.57	−0.023 %	19	0
DP (ours)	yes	406.58	35.58	0	**12**	0

design remains in the negligible-gain group.

7. Discussion and limitations

Design principles for connection benchmarks. The two coverage principles (magnitude families for unbalance, phase families for phase-recombining connections) are motivated by Prop. 2. We expect them to carry over to other machines: pattern families should be chosen by the span of the target connections' current vectors, not by sample count. Multi-phase and fault reconfiguration studies are the natural next testbeds, where the "phase" family is the default recommendation.

What structure buys. The co-energy network is best on the hard asym, delta and tap50 flux hold-outs and remains competitive on flux for tier-1 connections; the MLP retains a small in-support flux advantage. The torque channel no longer collapses after correcting the physical current scale: $R^2 \geq 0.981$ for the standard hold-outs and 0.834 for the open-phase case, while the local torque–flux residual is 0.67 %. Thus the same scalar potential now provides both accuracy and consistency for the downstream optimisation problem, rather than trading one for the other.

Limitations. (i) 2-D magnetostatics with a research-grade solver. The 64-point coarse/fine check gives flux correlation 0.997 and a 6 % norm-ratio difference, but torque correlation 0.751 ($R^2 = 0.377$). Pointwise torque labels remain mesh-sensitive even though the aggregate cycle and external checks are consistent. The FEMM check shows a 1.13× inductance normalisation factor, 2.1 % fine-mesh shape residual, 7.9 % PM fundamental error, 20.2 % mean nonlinear flux difference, and 7.5 % torque difference at the tested point. These limit absolute flux claims, while the 36-point end-to-end torque tests are stable ($R^2 \approx 0.9$). (ii) End effects, temperature, mechanical and changeover transients are excluded; the changeover circuit enters through a per-event energy, swept over four orders of magnitude (5 mJ–100 J). (iii) The drive-cycle study is a simulation case study on a laboratory-scale machine; absolute savings percentages are machine- and vehicle-specific, and we make no homologation claims. (iv) Iron loss is a Bertotti proxy calibrated to the assumed steel curve, not measured; the inverter is an idealised SiC model, and vehicle sizing is a case study rather than a hardware design. (v) External validity: the quantitative benchmark and the drive-cycle case study derive from a single machine; the two coverage *principles*, however, are replicated on a second machine (§5.6). All ground truth remains simulation-only, and no measurement validation was performed. A commercial FE cross-check is reported in Table 2; it exposes remaining discretisation/model-form differences but no qualitative zero-shot torque failure. Replication on a measured machine remains required before the design principles can be treated as general.

8. Conclusion

A winding connection is a linear constraint, not a new machine. Encoding that fact into the learning problem yields one surrogate queryable across winding connections, coverage-dependent zero-shot transfer, and exact physical consistency from a scalar co-energy head. It also yields a drive-cycle co-design loop that prices reconfiguration with full loss accounting—changeover energy, dwell constraints and system losses. On the WLTC class-1 case study, every fixed connection completes the cycle; the optimal schedule is 0.006 % less efficient than fixed 4S. Across 105 bus-voltage/gear/cycle combinations, no case requires reconfiguration to restore feasibility; 24 feasible cases gain at least 1 %, with a maximum of 2.77 %. Reconfiguration should therefore be justified by a specific measured constraint, not assumed to improve efficiency. The benchmark and its two data-design principles are, we believe, the most durable contributions: they tell the next study, whatever its machine, what data to collect and why. Future work will address geometry-parametric co-energy operators (design-space generalisation), thermal coupling, and hardware validation of the scheduling layer on a changeover test rig.

Reproducibility

All FE data (23,000+ samples across the benchmark and drive-cycle pools), the solver, the training and evaluation code, trained checkpoints, benchmark splits, the official WLTC class-1 trace preprocessing, the second-machine replication scripts, the FEMM cross-check and the 36-point unseen-connection tests, and one-command figure scripts are released at [[repository URL on acceptance]].

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